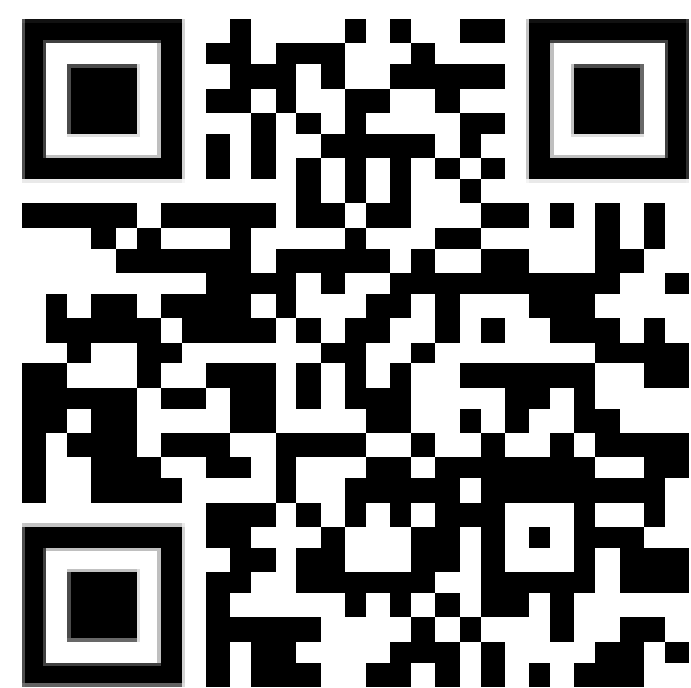




Intelligent Logistics - AI driven port logistics system

Bernardo Lázaro, Diogo Duarte, Paulo Cunha,
Rafael Ferreira, Tomás Hilário
Advisor: Prof. Daniel Corujo
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Background

To increase **efficiency** and **reduce costs**, port authorities are adopting Information and Communication Technologies (ICT) and Industry 4.0 concepts. With the democratization of **Artificial Intelligence**, **cloud computing**, and **process digitization**, innovative solutions are emerging to make logistics more **intelligent**, **automated** and **sustainable**.

AI-APP - Artificial Intelligence Domain

The **AI-APP** is a distributed, event-driven AI system composed of three containerized microservice agents forming a hazardous **vehicle detection pipeline** for truck detection, license plate reading and ADR hazard identification.



Fig 1 - Agents' Detection

V-APP - Validation and Operations Domain

The **V-APP** is the platform used by port operators to manage, monitor, and automate truck access and logistics.

- **Decision Engine/Infraction Engine** - Validates gate access and detects highway violations.
- **Data Module** - Manages port data and image storage with a polyglot architecture.
- **V-Brain** - Optimizes network resources dynamically for efficient operations.

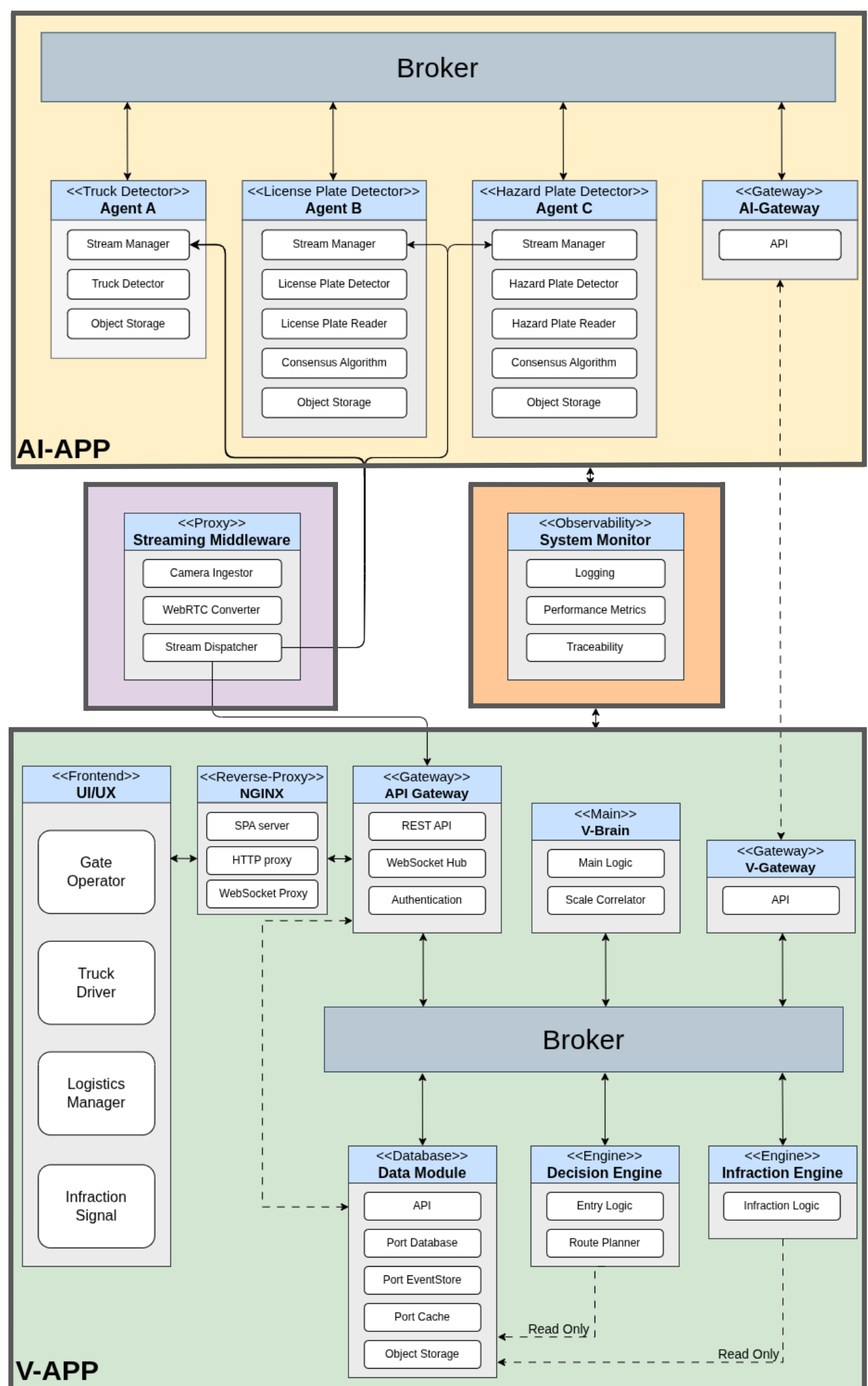


Fig 2 - System Architecture

Conclusion - Pilot Deployment

A **pilot trial** was conducted at the Port of Aveiro using a two-stage system with real cameras.

- 1.The first stage performs real-time truck and **ADR hazard detection**, triggering restriction signals whenever necessary.
- 2.The second stage simulates the port entry gate, completing **full vehicle validation** for access control.

The results showed improvements in both idle power consumption and truck waiting times at the port entrance.

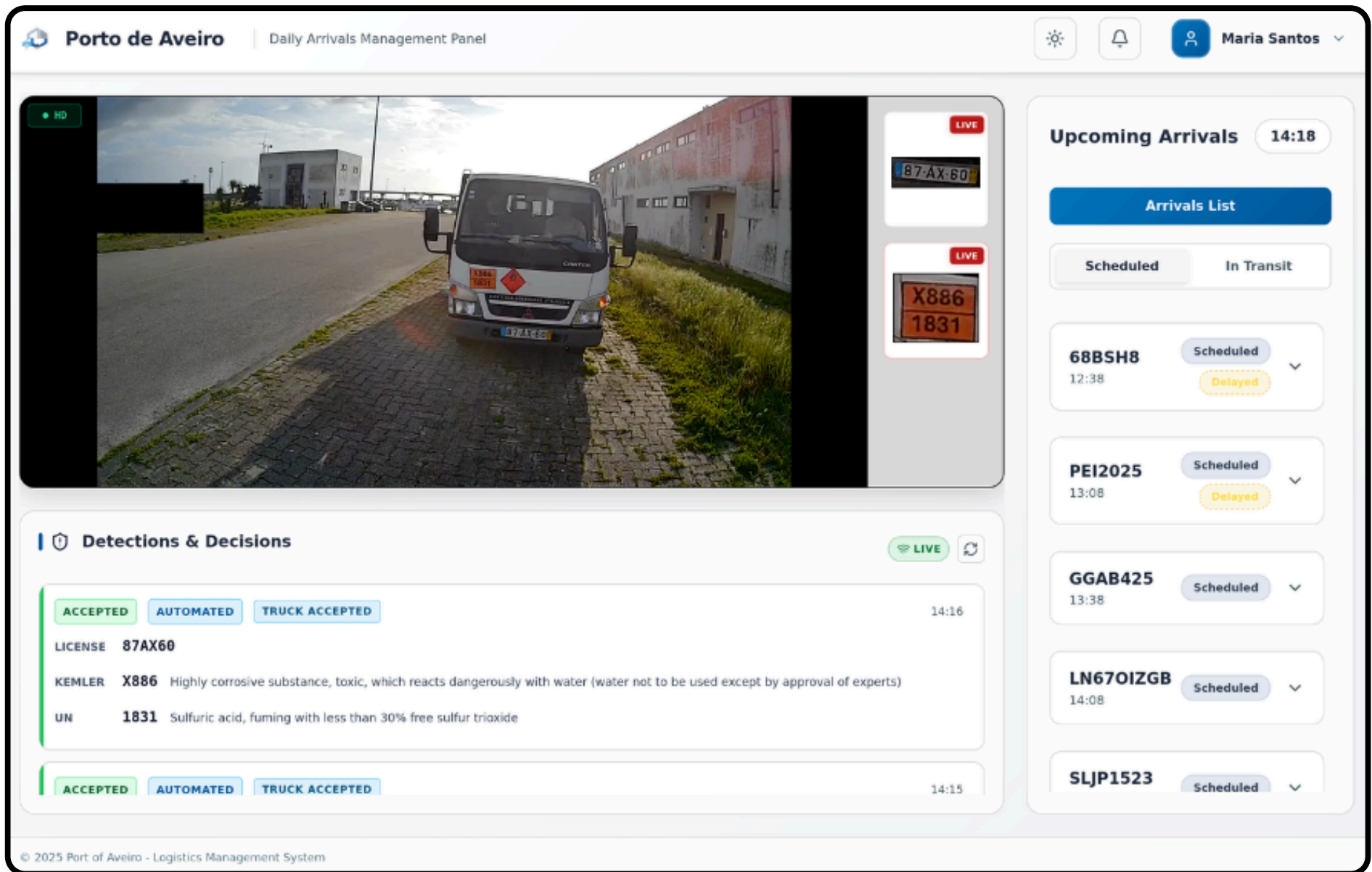


Fig 3 - Trial's Dashboard

